

Regulating AI in Healthcare Insurance: Closing the Gap Between Efficiency and Privacy

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I. Introduction

Over the past two decades, artificial intelligence has shifted from humanoid robots in science fiction movies to the operational backbone that defines coverage decisions for the largest health insurers in the United States. In May 2025, UnitedHealth Group reported nearly one thousand distinct AI use cases, supporting tasks within claims processing, risk scoring, fraud detection, prior authorization review, and predictive cost modeling. [20] These systems guarantee faster decisions, decreased administrative overhead, and ultimately, improved patient care. However, with the large amounts of data needed to train modern AI systems, are patients' privacy rights adequately being protected under existing U.S. health privacy regulations? The Health Insurance Portability and Accountability Act (HIPAA), the Health Information Technology for Economic and Clinical Health Act (HITECH), and the California Consumer Privacy Act and California Privacy Rights Act (CCPA/CPRA) were designed to protect identifiable health records, or protected health information (PHI) and govern how that data is stored, shared, and secured. However, they were not designed with modern-day advanced AI technologies in mind. These privacy regulations do not regulate the inferences AI systems generate from non-PHI data, the opacity of algorithmic denial rationales, or the use of predictive systems to determine who is insurable.

In this paper, I argue that efficiency and privacy in AI systems within health insurance is not inherently in conflict and requires novel regulatory frameworks developed with modern technology in mind. This paper proceeds in six parts. Section II examines how AI is used in health insurance operations, what data these systems consume, and how they generate sensitive inferences beyond the data they process. Section III identifies risks and concerns of AI. Section IV analyzes the limits of HIPAA, HITECH, and CCPA/CPRA in addressing these risks and identifies three structural privacy and safety gaps. Section V examines how the industry and state legislatures have begun responding, and where those responses fall short. Finally, section VI proposes a set of targeted reforms for efficient, private AI systems.

II. AI in Healthcare Insurance: Uses, Tasks, and Data Practices

To understand whether existing privacy regulations adequately protect patients from AI-related risks, it is important to understand what AI systems actually do within health insurance operations and why insurers have adopted them so rapidly. Insurance is an information-processing business that collects data on millions of members and makes coverage decisions based on historical records. However, these processes are extremely expensive. Administrative costs in the U.S. health insurance sector are estimated to account for roughly 25% to 34% of total healthcare spending. [7] For insurers operating on thin margins in a capitalist society, AI's promise of efficiency offers a powerful opportunity to reduce administrative overhead while simultaneously improving decision-making consistency and speed.

A. How AI Is Used: Current Applications in Insurance Operations

AI has been introduced to nearly every stage of the patient and provider experience. With AI, claims can be processed in seconds rather than days to reduce staffing costs, prior authorizations can be approved in minutes to improve care coordination, and suspicious activity can be detected across millions of claims simultaneously to identify fraud, saving insurance companies millions of dollars. [17] Claims processing and adjudication represent the highest-volume application of AI in insurance operations. Supervised learning systems, which are trained on millions of historical claims labeled as approved, denied, or fraudulent, learn to identify patterns in new claims that predict the appropriate outcome. [9] Risk scoring and predictive cost modeling use AI to predict which members are likely to incur high healthcare costs in the future. These predictions can help design benefit structures, allocate care management resources, and inform coverage pricing. [17] Fraud detection and anomaly identification use unsupervised learning systems to discover patterns in unlabeled data and detect unusual claim patterns. These systems scan billing records across thousands of providers, facilities, and member populations simultaneously, flagging outliers that may be easily missed by a human reviewer for further assessment. [9]

B. What Data These Systems Consume

Primary data sources used by health insurance systems include: (1) claims records that document patient medical encounters, diagnoses, and procedures, (2) pharmacy data, such as medication names, dosages, and refill histories, (3) laboratory results, (4) eligibility and enrollment records, and (4) provider network information. [6] However, as modern society digitizes and collects more data, modern AI systems in insurance often draw on sources beyond traditional clinical and administrative data. Insurers and their data partners have incorporated social determinants of health data, including information on housing stability, food security, educational attainment, and employment history, as well as commercial consumer

data purchased from third-party data brokers. This commercial data can include purchase histories, loyalty program records, fitness tracker outputs, magazine subscription lists, and financial transaction patterns. When combined, this clinical and commercial data creates an intimate profile that may disclose health-related information the individual never intended to share. [6]

C. Inferences Beyond the Data: How AI Generates Sensitive Information

When a predictive model is trained on a population's claims history and commercial data, it learns statistical relationships that enable it to make novel inferences about individuals or groups. A model might learn that members with certain purchase patterns, geographic locations, and prescription histories tend to incur high mental health treatment costs and then apply that learned relationship to predict the mental health risk of a new member who has never disclosed a mental health condition. The model outputs a risk score that encodes a sensitive inference about that individual's likely health status that they never consented to generate and may never know exists. [9] An AI system might also identify clusters of members who share characteristics associated with high future cost, generating group-level risk profiles that are then applied to all members in a cluster regardless of their individual circumstances. [17] From a privacy and equity standpoint, this population-level inference is particularly concerning because it can lead to consequential decisions about individuals based on the characteristics of groups they resemble rather than their own documented health history.

III. Risks and Concerns of AI in Health Insurance

Despite AI's promise of efficiency, every application carries technical, ethical, and social risks that have led to growing concern amongst patients, physicians, researchers, and policymakers. Most Americans do not trust healthcare systems to use AI responsibly. A nationally representative survey of 2,039 adults conducted by the National Opinion Research Center and published in JAMA Network Open found that 65.8% of respondents had low trust in their healthcare system's ability to use AI responsibly, and 57.7% lacked confidence that their providers would protect them against potential AI-related harm. [15] The primary predictor of trust in healthcare AI was an overall trust in the healthcare system, meaning that layering AI onto an already distrusted and confusing set of processes only amplifies hesitation. [13]

A technical concern with modern AI systems, particularly large language models and generative AI tools, is the risk of hallucination. A hallucination is when a model produces a confident output that is factually incorrect, misleading, or sometimes even entirely fabricated. Generative models operate as 'black boxes', producing outputs that may sound plausible and convincing even when they are completely incorrect.

Stanford researchers studying AI in prior authorization processes found that some insurance company staff could not explain in basic terms how generative AI produces outputs or were unaware that AI could be biased, yet expressed high confidence in their ability to use the tools effectively. [13]

Beyond the risk of individual errors, AI systems in health insurance raise deep concerns about systemic unfairness. Algorithmic bias refers to the tendency of AI systems to produce outcomes that are skewed against particular groups because the data they learn from reflects documented patterns of inequity. [21] If patients from lower-income zip codes historically received fewer specialist referrals, or patients from certain racial groups were prescribed less aggressive pain management, the model learns these patterns and reinforces racial, economic, and geographic disparities in healthcare access. [14] Despite current regulations that prevent protected characteristics, such as socioeconomic and demographic factors, from directly affecting coverage decisions, AI systems can inadvertently create proxy variables that are highly correlated with these factors. [17] This correlation is often indirect and embedded within a complex model, making a biased AI model undetectable without careful, disaggregated auditing of outputs. [9]

Minority and underserved communities are more likely to be negatively impacted by the invisible algorithmic unfairness due to historical biases embedded in the training data. These populations are more likely to face financial, geographic, and language barriers and experience discrimination that prevents them from accessing quality healthcare. A study by Obermeyer et al., published in *Science*, found that a widely used health risk algorithm systematically underestimated the healthcare needs of Black patients compared to white patients with equivalent health status. This occurred because the algorithm used healthcare cost as a proxy for health need, reflecting historical underutilization of care by Black patients due to systemic barriers, rather than lower actual need. [16] The implications that this introduces are especially concerning because underserved patients are more likely to be harmed by biased algorithms, less likely to have those biases detected or corrected, and least positioned to seek redress when errors occur. [17]

IV. The Limits of U.S. Health Privacy Regulations in the Age of AI

Despite the increased use of AI, existing U.S. health privacy laws do not adequately protect patients from the specific harms that AI systems introduce. In this section, we will explore the limitations of three primary regulatory frameworks overseeing health data privacy in the age of AI.

A. HIPAA: Outdated Protections

HIPAA serves as the foundation of health data privacy in the United States. Enacted in 1996, its Privacy Rule establishes standards for how covered entities, including health plans, healthcare clearinghouses, and most healthcare providers, may use and disclose individually identifiable health information, or Protected Health Information (PHI). [6] HIPAA's Security Rule extends these protections to electronic PHI, requiring covered entities to implement administrative, physical, and technical safeguards to prevent unauthorized access or disclosure. HIPAA also has a strict de-identification standard that requires covered entities to remove patient identifiers, such as name, address, date of birth, and Social Security number, prior to using or sharing health data. Once data satisfies this standard, it is no longer considered PHI and falls entirely outside HIPAA's protections.

For health insurers using AI, HIPAA creates a few enforceable obligations. Any AI vendor working with a covered entity must sign a Business Associate Agreement (BAA), which contractually extends HIPAA's privacy and security requirements to the vendor. [18] PHI used to train or operate an AI model must be handled in compliance with the Privacy Rule, and unauthorized disclosures can trigger significant civil and criminal penalties. However, HIPAA falls short in the new age of AI. Its outdated PHI definitions and de-identification standards, combined with a lack of requirements for AI transparency and appeal mechanisms, leave significant gaps in patient protection.

Currently, PHI includes information that is both individually identifiable and health-related. It does not regulate sensitive inferences or predictions derived from data. Modern AI systems can combine de-identified claims records, commercial consumer data, and geographic information to generate a risk score that reveals a lot about a person's likely health status. [4] However, under HIPAA's current definitions, this output would not qualify as protected PHI, meaning that it can be generated, stored, shared, and acted upon without triggering any of HIPAA's core protections. HIPAA's de-identification standard is also a cause for concern. This standard was robust in 1996, when combining de-identified records with other data sources to re-identify individuals was technically difficult. However, modern machine learning techniques can re-identify individuals from de-identified datasets with surprising accuracy. [9] Research has demonstrated that a small number of attributes, such as zip code, birthdate, or a rare diagnosis, can be sufficient to uniquely identify a person within a supposedly de-identified dataset. [19] Finally, HIPAA lacks a mechanism for patients to understand or challenge algorithmic decisions. The law gives individuals the right to access their health records and to request corrections to inaccurate information, but it does not require insurers to disclose that AI was involved in a coverage determination,

explain how a model reached its conclusion, or provide any accounting of the automated process that produced a denial. [4] A patient who receives a denial letter has no legal right under HIPAA to know that an algorithm generated it, what data trained that model, or the model's known error rate.

B. The HITECH Act: Better Enforcement, Same Limitations

The HITECH Act of 2009 enhanced HIPAA's enforcement framework by introducing tiered civil penalty structures based on the level of culpability involved in a violation, increasing the maximum penalties for willful neglect, extending direct HIPAA liability to business associates, and enhancing individuals' rights to access their electronic health records. [6] However, these improvements focused on making existing frameworks more enforceable rather than expanding the categories of information that HIPAA protects or creating new accountability mechanisms. HITECH essentially addresses the enforcement of rules that were not written with AI in mind.

C. CCPA/CPRA: A Step Forward, With Important Limits

California's Consumer Privacy Act, expanded by the California Privacy Rights Act, provides an updated framework for addressing AI-related privacy risks. The CCPA/CPRA grants California residents broad rights over their personal information, including the right to know what data is being collected, the right to request deletion, the right to opt out of the sale or sharing of their data, and the right to opt out of automated decision-making. [6] The CPRA also introduced the category of "sensitive personal information," including health and medical information, that requires greater protection. For health insurers operating in California, these provisions impose obligations beyond HIPAA, but significant limitations remain. Firstly, CCPA/CPRA only covers California residents. While more states are implementing laws modeled on the privacy regulations covered within CCPA/CPRA, there is currently no federal standard. This creates a preemption limitation, in which health information already regulated under HIPAA may require modifications or displacements of certain CCPA obligations. Another limitation is that CCPA/CPRA's automated decision-making provision does not require insurers to explain model logic or demonstrate that those models are free from bias. Additionally, in the context of health insurance, the right to opt out of automated decision-making may not always be feasible. Patients may not realize that AI is being used, and those who do often have no viable alternative if they want coverage.

D. The Three Gaps That Current Law Cannot Close

Taken together, HIPAA, HITECH, and the CCPA/CPRA leave three structural gaps that are directly relevant to how AI is used in health insurance today.

The first is the inference gap. Existing law protects identified health records but not the sensitive inferences that AI systems derive from non-PHI data. An insurer can purchase commercial consumer data, combine it with de-identified claims records, and generate a detailed health risk profile for every member, without any of this activity being subject to HIPAA and without the patient having any right to know it occurred. The most revealing and consequential outputs of AI systems in health insurance may never touch a single piece of PHI, yet they shape coverage decisions in ways invisible to both the patient and the regulator. [4][9]

The second is the transparency gap. None of the three frameworks requires insurers to disclose the role of AI in coverage decisions, explain the factors that drove an algorithmic recommendation, or notify patients when a denial was generated by a model rather than a human reviewer. This lack of transparency makes it impossible for patients to identify errors, for regulators to audit for discriminatory patterns, and for physicians to advocate effectively for their patients. [13]

The third is the coverage determination gap. When AI models are used to predict future risk and inform decisions about whether to offer coverage and at what cost, the systems are functioning as gatekeepers to medical care. The Affordable Care Act prohibits insurers from denying coverage or charging higher premiums based on pre-existing conditions, but this prohibition was not designed to address algorithmic profiling that uses socioeconomic and geographic proxies to assign risk scores that influence pricing or coverage terms, thereby inadvertently producing discriminatory effects without explicitly referencing any protected characteristic. [17] Current law does not clearly prohibit this practice, and none of the three frameworks described above require insurers to audit for it.

Health privacy laws have not yet adapted to the rapidly advancing technological landscape. Addressing these limitations will require new regulatory and technical frameworks that are specifically designed for the AI era.

V. Industry Responses: Mitigating Risk and Restoring Trust

With the increased use of AI came greater public criticism, legislative pressure, and scrutiny from physicians and legislators. The health insurance industry has begun responding with industry-led pledges and process reforms, legislative and regulatory interventions at the state level, and the adoption of technical privacy-enhancing tools.

A. Voluntary Industry Pledges and Process Reforms

The most visible industry response to criticisms of AI has been a series of voluntary commitments by major insurers. In June 2025, a coalition of major health insurers, including Aetna, Cigna, UnitedHealthcare, and Blue Cross Blue Shield plans, signed a pledge committing to a set of prior authorization reforms. [11] These commitments included reducing the overall volume of services subject to prior authorization by January 1, 2026, strengthening transparency on how decisions are communicated, and guaranteeing that all prior authorization denials based on medical necessity would be reviewed by a licensed and qualified clinician. [11] Some insurers have begun implementing these changes. Aetna has reported automating one in four prior authorization approvals for near-instant decisions, while Cigna has eliminated prior authorization requirements for nearly 100 services and committed to issuing decisions within one business day for at least 95 percent of electronic requests by January 1, 2026. [8] Despite these improvements, little has changed in practice for patients and clinicians, and the inference and transparency gaps remain unaddressed. [8] Without an enforcement mechanism or judicial penalties for insurers that fail to meet their commitments, it raises questions about whether these pledges can bring about legitimate, lasting change or if they are just a public relations stunt to build public trust.

B. Legislative Responses: States Lead Where Federal Law Falls Short

In the absence of federal legislation addressing AI in health insurance, states have taken measures to enforce their own regulatory frameworks. A major legislative development is California's Senate Bill 1120, the Physicians Make Decisions Act, which took effect January 1, 2025. [3] The law requires that any denial, delay, or modification of healthcare service based on medical necessity be made by a licensed physician or other healthcare professional competent to evaluate the specific clinical issues involved, rather than solely by an algorithm. [3] AI tools may still be used to supplement the review process, but they cannot replace the clinician's final determination. SB 1120 targets the population-level inference problem by prohibiting AI systems from basing prior authorization decisions solely on group-level datasets. It also requires these systems to comply with existing medical privacy laws, be open to inspection, and not discriminate directly or indirectly on the basis of protected characteristics. [3]

At the federal level, the Centers for Medicare & Medicaid Services issued a rule in 2024 clarifying that Medicare Advantage organizations have to ensure that medical necessity determinations are based on the specific individual's circumstances and cannot rely on algorithms or software that do not account for those

circumstances. [10] Although these state and federal developments are a step forward, they remain limited by state-specific constraints and are narrowly focused on prior authorizations. They do not address the inference gap, the use of commercial data in risk scoring, or the use of AI in determining initial coverage eligibility.

C. Technical Responses: Privacy-Enhancing Technologies

The industry has also begun exploring technical tools designed to improve privacy protections. Differential privacy (DP), a mathematical model that adds carefully calibrated statistical noise to datasets or model outputs, makes it difficult to infer information about any individual from the system's outputs. [9] Several major technology companies use differential privacy in consumer-facing products, and it is actively being explored in healthcare AI research. [9] A 2025 scoping review of differential privacy applications in medical deep learning found that DP can maintain clinically acceptable model performance under moderate privacy settings, particularly in imaging tasks, while providing meaningful protection against re-identification. [2] However, this study also found that under strict privacy requirements, model accuracy can decline significantly. This decline is often largest for smaller or underrepresented subgroups in the training data, which are already at high risk of algorithmic bias. While differential privacy can be used as a valuable tool, it is not a complete solution and must be implemented cautiously to avoid inadvertently worsening equity outcomes.

Another approach that is gaining traction is federated learning. Rather than centralizing patient data in a single location for model training, federated learning allows AI models to be trained across multiple institutions' data without that data ever leaving its source. Each institution trains the model locally on its own data and shares only the updated model. [12] This approach lowers the risk of large-scale data breaches and limits the exposure of sensitive patient information during model development.

Explainable AI (XAI) methods are also being explored to make the outputs of complex models more interpretable to humans. Explainable AI tools can generate plain-language summaries of the factors that contributed to a particular prediction or recommendation, enabling clinicians and appeals reviewers to evaluate the model's reasoning. [13] However, for the deep learning systems most commonly used in insurance applications, the explanations generated can be misleading and are often approximations rather than true representations of the model's internal logic. Requiring explainability without also requiring accuracy guarantees for those explanations risks can create a false sense of transparency.

Taken together, these technical responses can lead to the development of AI that values accountability and privacy. However, none of them are currently required by law, resulting in uneven adoption across the industry.

VI. Policy and Technical Recommendations for Responsible AI Use

Given its rising popularity, corporate interest, and promise for efficiency, the use of AI in healthcare feels inevitable. The question is no longer whether AI should be used, but under what constraints AI should operate and how fairness, transparency, and patient autonomy can be ensured. This section argues that AI can be used safely, privately, and equitably in health insurance, but only with a set of strict, enforceable reforms that address the inference, transparency, and coverage determination gaps.

A. Require Explainability for Denial Decisions and Auditability of Models

The AMA has specifically called for oversight of payers' use of AI in medical necessity determinations as a top regulatory priority, with nearly half of physicians surveyed ranking it among their top three concerns. [1] Insurers using AI in claims adjudication and prior authorization review should be required to disclose when AI was involved in making a determination, provide a meaningful explanation of the factors that contributed to that determination, and make those explanations available to patients, clinicians, and appeals reviewers acting on the patient's behalf. Although California's SB 1120 already moves in this direction for prior authorization decisions and the European Union's AI Act requires disclosure and human oversight for high-risk AI systems used in insurance contexts, [3] a federal standard would broaden these protections to patients in all states. To assess for discriminatory disparities through proxy variables and biased training data, explainability requirements must be paired with mandatory third-party auditing of model outputs. These disparities are often invisible without careful analysis of disaggregated outputs and would require insurers to conduct and publicly report regular audits broken down by race, income, gender, geography, and other relevant characteristics. For opaque model architectures, such as deep learning systems, insurers should be required to either use models whose reasoning can be meaningfully explained or to demonstrate that their model's outputs meet fairness and accuracy standards through rigorous testing, even if its internal logic cannot be fully realized. When neither condition can be met, the use of that model in high-stakes coverage decisions should be prohibited. This would create both an accountability mechanism and an incentive to detect and correct bias before it causes harm.

B. Limit the Use of Non-Medical Data in Coverage Decisions

One of the most concrete steps regulators could take to address both the inference gap and the proxy discrimination problem is to limit the categories of data that insurers may use as inputs to AI systems that inform coverage decisions. Currently, there is no federal restriction on insurers purchasing and incorporating commercial consumer data into risk-scoring or prior-authorization models. This includes purchasing histories, location data, fitness tracker outputs, and financial transaction records that can disclose sensitive health information the patient never intentionally disclosed and can serve as a proxy for race or income in ways that produce discriminatory outcomes while evading anti-discrimination law. [6]

A targeted federal regulation should prohibit the use of non-medical commercial data in AI models that inform coverage eligibility, premium pricing, or medical necessity determinations. Permissible inputs should be limited to data directly relevant to the individual's health status and clinical circumstances, rather than behavioral or commercial data purchased from third-party brokers. This restriction would directly address the inference gap by limiting the raw material from which sensitive health inferences can be generated without the patient's knowledge or consent. This recommendation is consistent with Colorado's External Consumer Data and Information Sources regulations, which require insurers to report how they use commercial data sources in AI models and to develop governance frameworks overseeing that use. [10]

C. Strengthen Rights to Appeal Automated Decisions

The existing appeals process for insurance coverage denials was designed for a time when decisions were made by humans. When a clinician reviewer denies a claim, the patient or their physician can challenge the reasoning, present additional evidence, and request an external review. When an AI system generates a denial, the patient may not know AI was involved, cannot access the model's reasoning, and cannot determine whether the denial reflects a hallucination, bias in the training data, or the correct application of clinical criteria. At a minimum, federal law should require that any adverse coverage determination generated or substantially informed by an AI system include: (1) a clear disclosure that AI was used and a plain-language description of its role; (2) the right to request human review of the AI's recommendation by a licensed clinician; and (3) an expedited external review pathway for cases where the patient alleges that the AI system produced a discriminatory or factually incorrect result. These requirements build on what California's SB 1120 has already established for prior authorization and extend it to all AI-informed coverage determinations. [3]

Strengthening appeal rights also requires tackling the practical barriers that prevent patients from using the rights they already have. Studies consistently show that appeal rates for insurance denials are very low, and that denials overturned on appeal represent a large share of all denials, meaning that many valid claims are being denied and then abandoned rather than successfully challenged. [13] Underserved patients who lack the time, health literacy, or administrative support to navigate the complex and lengthy appeals process are often placed in a position where they cannot realistically advocate for themselves or pursue an appeal, even when they urgently need care. A well-designed appeals system for AI-driven decisions should include simplified procedures, access to patient advocates, and defined timelines.

D. Introduce Inference Protections, Not Just Data Protections

Inference protections should be introduced as legal safeguards that apply to the sensitive outputs of AI systems. Any AI-generated output that constitutes a health-related inference about an individual, including risk scores, coverage eligibility assessments, and predictive diagnoses, should be subject to privacy and security protections equivalent to those afforded to PHI under HIPAA, regardless of whether PHI was used to generate it. This would mean that such inferences could not be generated without appropriate authorization, could not be disclosed to third parties without consent, would be subject to patients' rights of access and correction, and would have to be disclosed to the patient if used in a coverage decision. [4] HIPAA's de-identification standard should also be updated to require covered entities to conduct formal re-identification risk assessments using current statistical methods before using de-identified datasets for AI training or operations. Where re-identification risk exceeds defined thresholds, entities should be mandated to implement privacy-enhancing technologies such as differential privacy to bring that risk within acceptable bounds. [19]

VII. Conclusion: A Path Forward

The integration of AI into health insurance operations has moved faster than the regulatory frameworks designed to protect patients. As this paper has shown, HIPAA, HITECH, and CCPA/CPRA were built for a different technological era. These frameworks protect identifiable health records, but they do not protect the sensitive inferences AI systems generate from commercial data, the opaque models that produce uninterpretable denial rationales, and the predictive systems quietly functioning as gatekeepers that define who is worthy of receiving medical care. These risks are not evenly distributed, and minority and underserved communities are the most likely to be harmed by biased training data, the least likely to have those harms detected, and the least positioned to navigate the appeals processes that exist on paper but rarely work in practice. When AI encodes historical inequity and applies it at scale, the consequences fall

heaviest on the people the healthcare system has already failed. Despite these challenges, efficiency and privacy can be complementary goals, achievable with the right technical tools and regulatory frameworks. The reforms proposed in this paper, inference protections, limits on commercial data use, mandatory explainability and auditing, strengthened appeal rights, and a requirement of robust security practices, including differential privacy and federated learning, pave the way towards reliable, safe, and efficient AI systems in health insurance.

References

- [1] American Medical Association. (2024). 2024 AMA Prior Authorization Survey. AMA. <https://www.ama-assn.org/system/files/prior-authorization-survey.pdf>
- [2] Arxiv / PMC. (2025). Differential privacy for medical deep learning: methods, tradeoffs, and deployment implications. <https://arxiv.org/pdf/2506.00660>
- [3] California Senate Bill 1120 (2024). The Physicians Make Decisions Act. Signed into law September 28, 2024. <https://sd13.senate.ca.gov/news/press-release/september-30-2024/governor-signs-physicians-make-decisions-act-keeping-medical>
- [4] Censinet, Inc. (n.d.). Is your AI HIPAA-compliant? Why that question misses the point. Censinet. <https://censinet.com/perspectives/is-your-ai-hipaa-compliant-why-that-question-misses-the-point>
- [5] Duff-Brown, B. (2026, January 6). When AI algorithms decide whether your insurance will cover your care. Stanford Health Policy. <https://healthpolicy.fsi.stanford.edu/news/when-ai-algorithms-decide-whether-your-insurance-will-cover-your-care>
- [6] Eagle, R. (2026, February 2). AI, medical data, and privacy: How personal information is protected. Healthline. <https://www.healthline.com/health/drugs/ai-your-medical-data-and-privacy>
- [7] Himmelstein, D. U., Lawless, R. M., Thorne, D., Foohey, P., & Woolhandler, S. (2019). Medical Bankruptcy: Still Common Despite the Affordable Care Act. American Journal of Public Health, 109(3), 431–433. <https://doi.org/10.2105/AJPH.2018.304901>
- [8] iwins.com. (2025, December 11). Health insurers slowly cut prior authorizations. <https://www.iwins.com/blog/health-insurers-slowly-cut-prior-authorizations/>
- [9] Jangid, P., Meena, A., Prerna, R., Hashmi, Z. H., Kumar, V., & Bhagtani, H. (2025). The Role of Artificial Intelligence in Safeguarding Patient Privacy in Healthcare Systems. Journal of pharmacy & bioallied sciences, 17(Suppl 2), S1083–S1085. https://doi.org/10.4103/jpbs.jpbs_381_25
- [10] Kansas Legislative Research Department. (2026, March 2). Briefing Book 2026: Artificial Intelligence Use in Health Insurance. <https://klrd.gov/2026/03/02/briefing-book-2026-artificial-intelligence-use-in-health-insurance/>
- [11] Managed Healthcare Executive. (2025, June 25). Plans agree to prior auth reform. Physicians and business groups want measurable results. <https://www.managedhealthcareexecutive.com/view/plans-agree-to-prior-auth-reform-physicians-and-business-groups-want-measurable-results>

- [12] MedTech Solutions. (2025, December 10). HIPAA and AI in healthcare: Lessons from 2025 and what's coming in 2026.
<https://medtechsolutions.com/resource-center/blog/hipaa-and-ai-in-healthcare-lessons-from-2025-and-whats-coming-in-2026/>
- [13] Mello, M. M., Trotsyuk, A. A., Djiberou Mahamadou, A. J., & Char, D. (2025). The AI arms race in health insurance utilization review: promises of efficiency and risks of supercharged flaws. *Health Affairs*. <https://doi.org/10.1377/hlthaff.2025.00897>
- [14] Nazerali, R., et al. (2023). Data privacy in healthcare: In the era of artificial intelligence. *PMC / National Library of Medicine*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10718098/>
- [15] Nong, P., & Platt, J. (2025). Trust in healthcare systems to use artificial intelligence responsibly. *JAMA Network Open*. Published February 2025. (Reported via: University of Minnesota School of Public Health.
<https://www.sph.umn.edu/news/most-people-do-not-trust-healthcare-systems-to-use-artificial-intelligence-ai-responsibly/>)
- [16] Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453.
<https://doi.org/10.1126/science.aax2342>
- [17] Ramezani-A, M., Bakhtiari, A., Mobinizadeh, M., Daroudi, R., Rabiee, H. R., Olyaeemanesh, A., Fazaeli, A. A., Mostafavi, H., Ramezani-B, M., Sazgarnejad, S., Bordbar, S., & Takian, A. (2025). Artificial intelligence applications in health insurances: a scoping review. *Cost effectiveness and resource allocation* : C/E, 23(1), 53. <https://doi.org/10.1186/s12962-025-00640-w>
- [18] Rezaeikhonakdar D. (2023). AI Chatbots and Challenges of HIPAA Compliance for AI Developers and Vendors. *The Journal of law, medicine & ethics : a journal of the American Society of Law, Medicine & Ethics*, 51(4), 988–995. <https://doi.org/10.1017/jme.2024.15>
- [19] Sweeney, L. (2000). Simple demographics often identify people uniquely. *Carnegie Mellon University, Data Privacy Working Paper 3*. <https://dataprivacylab.org/projects/identifiability/paper1.pdf>
- [20] Wolter, S. (2025, May 5). UnitedHealth now has 1,000 AI use cases, including in claims. *The Wall Street Journal*.
https://www.wsj.com/articles/unitedhealth-now-has-1-000-ai-use-cases-including-in-claims-f3387ca3?mod=tech_lead_story
- [21] What Are The Social Implications Of AI?. *AI Tech-Wise*
<https://aitechwise.co.uk/what-are-the-social-implications-of-ai/>